

Batch emissions report

exp2_rag_2026-09-11_00-53-52

Models: Unknown

Pipeline: RAG | Questions: 2 | Successful: 2 | Failed/unknown: 0

Batch runtime: 158.01 s | Successful requests/min: 0.75946

Measured cost (all requests)	Observed sum	Valid / all
Energy (Wh)	1.0615	2 / 2
Emissions (g CO2e)	0.75734	2 / 2

Method and data quality

Statistics and distributions use successful requests with finite measurements. Cost totals include measured failed requests. Missing/non-finite values are excluded, never replaced with zero. Partial totals are not estimates of unmeasured cost.

Variance and standard deviation use the sample definition ($n - 1$); both are N/A for fewer than two observations. Variance has squared units. P95 uses linear interpolation. Small values use significant digits or scientific notation.

Historical pipelines may already have stored missing telemetry as zero. This report cannot distinguish those defaults from measured zero. RAM fields describe estimated energy, not memory usage. Per-question charts use CSV row order; red points mark failed/unknown requests.

Metric	Missing	Zero
Latency (s)	0	0
Energy (Wh)	0	0
Emissions (g CO2e)	0	0
Retrieval latency (s)	0	0
Retrieval energy (Wh)	0	0
Retrieval emissions (g CO2e)	0	0
Generation latency (s)	0	0
Generation energy (Wh)	0	0
Generation emissions (g CO2e)	0	0
CPU energy (Wh)	0	0
GPU energy (Wh)	0	2
RAM energy (Wh)	0	0

Measurement scope and reliability

UNVERIFIED: This batch has no runtime measurement metadata. Its historical hardware coverage and backend methods cannot be established. Earlier zero defaults cannot be recovered.

A partial observed sum excludes unavailable readings. A complete batch cost cannot be claimed when coverage is incomplete. Zero readings are retained; unknown hardware coverage must not be interpreted as zero consumption.

Statistical summary

Latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	75.85	75.85	18.381	4.2873	72.819	78.882	78.579

Energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.53077	0.53077	0.00077054	0.027759	0.51114	0.5504	0.54843

Emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.37867	0.37867	0.0003922	0.019804	0.36467	0.39267	0.39127

Retrieval latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.04484	0.04484	0.00039942	0.019986	0.030708	0.058972	0.057559

Retrieval energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.00018233	0.00018233	6.2462e-08	0.00024992	5.6047e-06	0.00035905	0.00034138

Retrieval emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.00013008	0.00013008	3.1793e-08	0.00017831	3.9986e-06	0.00025616	0.00024355

Generation latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	75.806	75.806	18.553	4.3073	72.76	78.851	78.547

Generation energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.53058	0.53058	0.00078448	0.028009	0.51078	0.55039	0.54841

Generation emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.37854	0.37854	0.0003993	0.019982	0.36441	0.39267	0.39126

CPU energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.32027	0.32027	0.00025418	0.015943	0.309	0.33155	0.33042

GPU energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0	0	0	0	0	0	0

RAM energy (Wh)

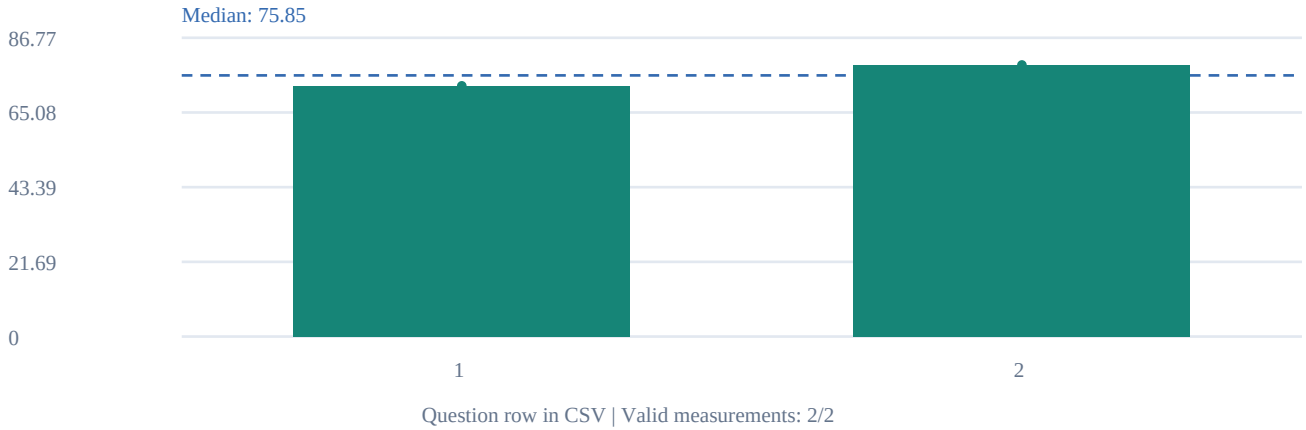
Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.21049	0.21049	0.00013961	0.011816	0.20214	0.21885	0.21801

Which questions cost the most?

Bars show individual requests; large batches use points to avoid overcrowding. Missing measurements are marked X, not shown as zero.

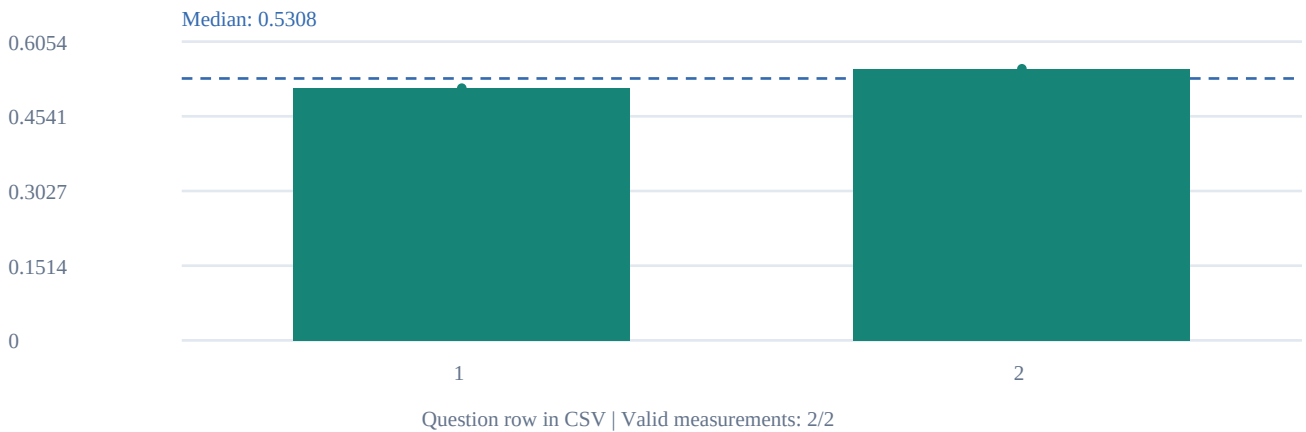
Latency (s)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median



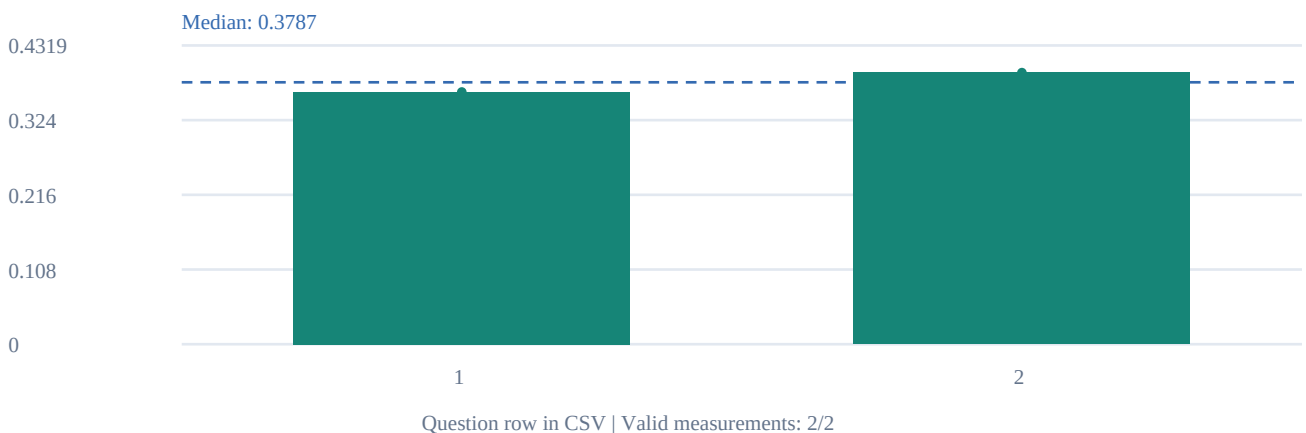
Energy (Wh)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median



Emissions (g CO2e)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median

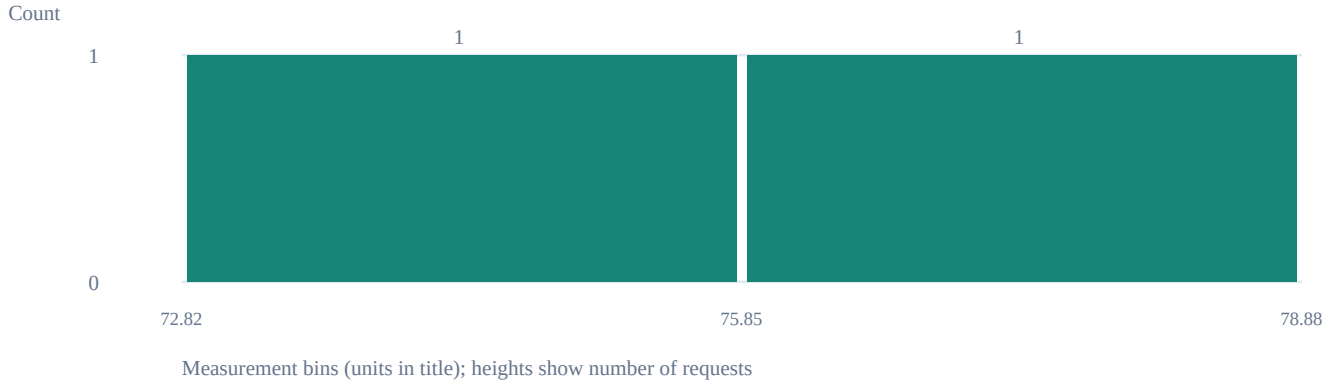


How much do requests vary?

Successful requests only. Bin boundaries show measurement values; bar heights and labels show request counts.

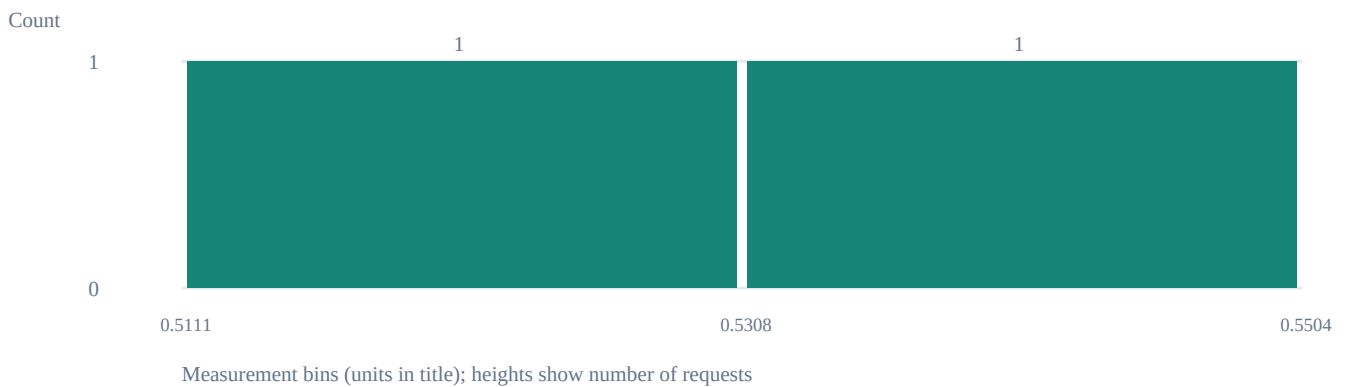
Latency (s)

Successful requests: n=2 | Mean 75.85 | Median 75.85



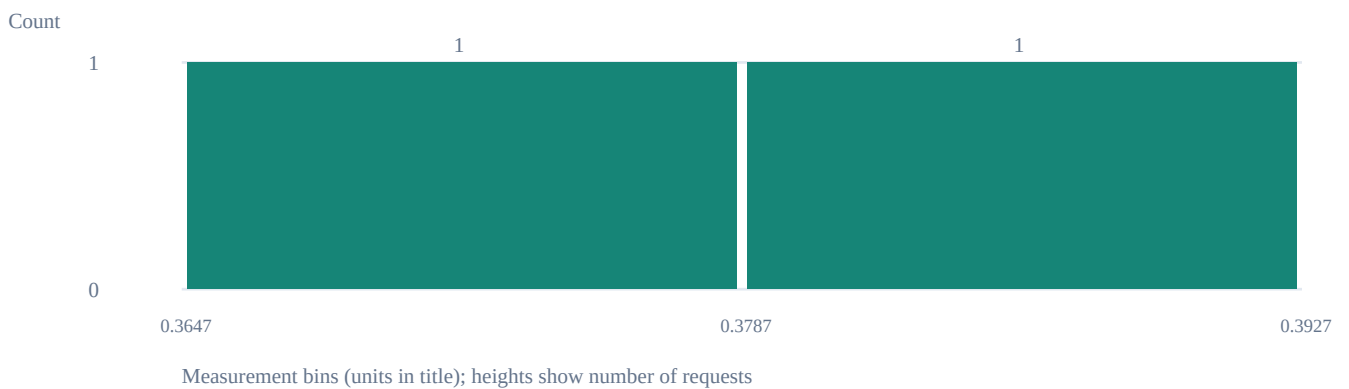
Energy (Wh)

Successful requests: n=2 | Mean 0.5308 | Median 0.5308



Emissions (g CO2e)

Successful requests: n=2 | Mean 0.3787 | Median 0.3787



Where is energy spent?

Compare observed component costs below. Amber bars indicate incomplete coverage; comparisons with different coverage may be misleading. These are tracking estimates, not isolated model or wall-socket measurements.

Hardware energy (Wh)

Observed sums across all requests; partial coverage is labelled.



Retrieval versus generation

Time (s)

Observed sums across all requests; partial coverage is labelled.



Energy (Wh)

Observed sums across all requests; partial coverage is labelled.



Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

